

CLAE: A Compact 12.5 Hz Bengali Speech Representation Encoder

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Abstract

We study a compact representation encoder for 16 kHz Bengali speech. Its 15.29M-parameter frontend and encoder produce 256-dimensional features z at 12.5 frames/s for downstream frozen-feature discrimination tasks; the full training model has 23.80M parameters. The 12.5 Hz resolution aligns with low-frame-rate neural audio codecs and tokenizers such as Mimi, DualCodec, and NanoCodec (Défossez et al., 2024; Li et al., 2025; Casanova et al., 2025), while CLAE remains continuous: its floating-point latent is neither quantized nor entropy-coded, so its scalar rate is not bitrate-comparable. A separate per-frame projector maps z to a 64-dimensional space p . Training uses waveform reconstruction as an auxiliary self-supervised pretext and information-retention objective, non-contrastive global/local view consistency in p to encourage nuisance-invariant features, and Variance-Invariance-Sketching Regularization (VISReg) in p to support non-collapsed, well-distributed representations. At the selected 210k-step checkpoint, frozen z features support the reported emotion, age, gender, and speaker probes; reconstruction on 400 Common Voice Bengali validation-manifest clips gives MR-STFT 0.747, STOI 0.850, ESTOI 0.762, and PESQ-WB 2.146. A matched 25k-step study shows that removing reconstruction worsens reconstruction metrics and two temporal-emotion probes in this setting. The main comparisons use one seed and objective reconstruction metrics; broader evaluation is needed to establish generality.

1 Introduction

Compact speech encoders are useful when their frozen features retain information for several downstream discrimination tasks without requiring task-specific encoder adaptation. This frozen-feature

framing follows speech representation work such as wav2vec 2.0 and HuBERT (Baeovski et al., 2020; Hsu et al., 2021) and the lightweight-head evaluation protocol of SUPERB (Yang et al., 2021). Modern neural audio codecs and tokenizers also commonly operate at low temporal resolutions, making 12.5 Hz a practical design point for a compact speech representation; DualCodec and NanoCodec, for example, report 12.5 Hz codec configurations (Li et al., 2025; Casanova et al., 2025). This does not make the systems bitrate- or architecture-equivalent. CLAE is continuous rather than a codec: its floating-point latent is neither quantized nor entropy-coded, so its scalar rate is not directly comparable to codec bitrate.

We present CLAE, a 23.80M-parameter continuous-latent training model trained on an approximately 2,000-hour inventory assembled from four Bengali speech sources. Its 15.29M-parameter frontend and encoder produce a 256-dimensional frame-level representation z at 12.5 Hz; downstream probes keep this feature extractor frozen and consume z . A distinct 64-dimensional projector output p is used only by two representation objectives: direct global/local view consistency and Variance-Invariance-Sketching Regularization (VISReg; Wu et al., 2026). View consistency encourages features stable to the augmentation and masking nuisances used to construct the views. VISReg supports a centered, scaled, and distributed projector population, countering degenerate representations. Reconstruction is an auxiliary self-supervised pretext and information-retention objective: the decoder generates a waveform from a corrupted copy of z , and the generated waveform is compared with the clean source in an 80-bin log-mel domain.

The contribution is empirical and implementation-specific. We provide: (i) a compact continuous 12.5 Hz Bengali speech repre-

sensation encoder evaluated through frozen-feature discrimination probes; (ii) a training objective that combines reconstruction for self-supervised information retention, non-contrastive view consistency for nuisance invariance, and frame-level VISReg for anti-collapse and distribution regularization; (iii) a Conformer-derived encoder with a project adaptation of Manifold-Constrained Hyper-Connections (mHC); and (iv) evaluations of the selected 210k-step checkpoint together with a matched 25k-step ablation suite. We assess the combined design through frozen-feature probes and ablations; the individual architectural and objective components build on prior work.

2 Related Work

Joint-embedding learning. Image JEPA introduced representation-space prediction with distinct context and target pathways (Assran et al., 2023); A-JEPA adapted masked-target JEPA ideas to audio (Fei et al., 2023). LeJEPA instead studies direct view agreement with isotropic-Gaussian regularization (Balestrierio and LeCun, 2025). CLAE uses a JEPA-inspired, non-contrastive per-frame view-agreement loss, but it is not canonical predictive JEPA: it has one shared encoder and no target encoder, exponential moving average, learned predictor, stop-gradient target path, masked-region prediction, or negative pairs. It also differs from LeJEPA in domain, waveform view construction, grouped consistency loss, frame-level population, supporting regularizer, and reconstruction.

Distribution regularization. VICReg combines invariance, per-dimension variance control, and covariance regularization (Bardes et al., 2022). VISReg retains explicit center and scale terms but replaces covariance control with random one-dimensional projections whose sorted samples are matched to Gaussian quantiles (Wu et al., 2026). We use VISReg to regularize the distribution of projector features and discourage collapse. Our frame-level adaptation is described in Section 3.3.

Continuous and discrete audio representations. Continuous Audio Language Models use a continuous variational bottleneck, KL regularization, additional speech distillation, and generative next-frame modeling (Rouard et al., 2025); those components are not used here. UniWav studies joint representation and generation objectives for speech (Liu et al., 2025a). By contrast, UniAudio and

UniTok-Audio illustrate autoregressive generation over discrete audio tokens (Yang et al., 2024; Liu et al., 2025b). Neural codecs such as SoundStream and EnCodec learn quantized, bitrate-controlled representations (Zeghidour et al., 2021; Défossez et al., 2023). Mimi, introduced with Moshi, is a separately pretrained common-evaluation reconstruction reference in our 25k study (Défossez et al., 2024). CLAE’s continuous floating-point latent is neither quantized nor entropy-coded, so its scalar rate is not directly comparable to codec bitrate.

Sequence encoder and residual connections. The encoder builds on the Conformer macaron block (Gulati et al., 2020). It incorporates squeeze-and-excitation but does not adopt the full down-sampling and local-attention design of FastConformer (Rekesh et al., 2023). Selected layers use a project adaptation of Manifold-Constrained Hyper-Connections (Xie et al., 2025).

3 Method

3.1 Architecture

Figure 1 summarizes the architecture. For a 16 kHz mono waveform x , the convolutional frontend has five Conv1D stages with channels [128, 256, 384, 512, 512], kernels [10, 8, 8, 4, 4], and strides [5, 4, 4, 4, 4], with GroupNorm and GELU. Its total stride is 1,280 samples, yielding approximately 12.5 frames/s. An eight-layer, 256-dimensional Conformer-derived encoder uses eight attention heads, a 1,024-dimensional feed-forward sublayer, rotary full self-attention, macaron feed-forward modules, a 9-tap convolution branch, dropout 0.1, and squeeze-and-excitation. It produces

$$z = E(F(x)), \quad z \in \mathbb{R}^{T' \times 256}. \quad (1)$$

A per-frame MLP projector with dimensions $256 \rightarrow 512 \rightarrow 64$ produces $p = P(z)$. The decoder and downstream frozen-feature probes use z , whereas the consistency and VISReg objectives use p .

The FiLM-conditioned waveform decoder starts with 768 channels, reverses the frontend with strides [4, 4, 4, 4, 5], and uses two residual blocks per upsampling stage with dilations [1, 3, 9]. The full training model has 23,803,125 parameters. The frontend, encoder, projector, and decoder contain 2,890,240, 12,403,220, 165,440, and 8,344,225 parameters, respectively. The frozen

frontend–encoder feature extractor therefore contains 15,293,460 parameters.

3.2 Waveform views and consistency objective

Training first samples one three-second source segment, and every view is derived from that same pre-cropped waveform. Two global views receive independently sampled waveform augmentation. Four local views remain full-length three-second waveforms but receive stronger waveform augmentation and waveform-aligned span masking; their frontend features can receive an additional independent frame mask. The waveform augmentations are gain, additive noise, centered low-pass filtering, and clipping only: they do not shift, speed-perturb, resample, independently crop, or temporally warp the signal. Waveform masks are constructed on the 1,280-sample frontend-frame grid and expanded onto the corresponding waveform samples; the optional frontend mask is also defined on that frame grid. Thus, frame index t refers to the same time interval across views, although its waveform content may be augmented or masked. All six views pass through the same frontend, encoder, and projector.

Let p_g and p_l denote projector features grouped into global and local views. For each example and frame, the center $\bar{p}_g$ is the mean of the two global projector features. The implemented consistency loss is

$$\mathcal{L}_{\text{view}} = \text{mean}[(p_g - \bar{p}_g)^2] + \text{mean}[(p_l - \bar{p}_g)^2]. \quad (2)$$

The two group means are added, so global and local groups have equal aggregate contribution despite containing two and four views. This JEPA-inspired objective is non-contrastive per-frame global/local agreement, not canonical context-to-target or masked-region prediction, and it is not a direct utterance-identity or class-discrimination loss. It encourages invariance to the specified augmentation and masking nuisances, which can be useful for the downstream discrimination probes, but those probes are separate evaluations of z .

3.3 VISReg adaptation

Figure 2 outlines the regularizer. The supporting VISReg implementation flattens projector frames from all views and examples into one population; it does not pool them. Under distributed training, the population is gathered across workers. Its effective shape is $N = 1$, $B = T'VB_{\text{global}}$, and $D = 64$. For population points $q_i \in \mathbb{R}^D$, it penal-

izes the population mean, centers the points, computes a per-dimension RMS scale with division by $\sqrt{B}$, and penalizes deviation from unit scale. Normalization uses a detached scale. On every forward pass, 256 normalized Gaussian directions are sampled. Values projected onto each direction are sorted and matched to standard-normal quantiles $\Phi^{-1}(i/(B+1))$. Center, scale, and projected-distribution shape terms have equal internal weight.

This implements the core VISReg center/scale/shape construction with project-specific frame-level and distributed adaptations. Our adaptation gathers frames across workers, adds 10^{-6} to the scale for numerical stability, assigns equal weights to the three terms, and casts target quantiles to the activation dtype. Training uses BF16 precision.

3.4 Reconstruction and total objective

The decoder receives z from the first global view. Before decoding, an independent span mask removes 50% of frames in spans of 4–24 frames and Gaussian noise with standard deviation 0.03 is added. The target remains the clean source waveform. The decoder outputs a waveform, which is compared with the clean waveform using an 80-bin log-mel loss with FFT size 1,024, window length 1,024, hop length 256, a Hann window, magnitude weight 1.0, log-magnitude weight 1.0, and spectral-convergence weight 0. Multi-resolution STFT is used only for evaluation.

The selected objective is

$$\mathcal{L}_{\text{total}} = 1.0\mathcal{L}_{\text{mel}} + 0.3\mathcal{L}_{\text{view}} + 0.7\mathcal{L}_{\text{VISReg}}. \quad (3)$$

The three coefficients weight losses with different numerical scales. Training uses no quantization, variational bottleneck, or adversarial objective.

3.5 mHC adaptation

At zero-indexed encoder layers 2 and 5, mHC widens the residual path to two streams. It applies learned static pre/post mixing and a Sinkhorn-constrained residual matrix with identity blending, uses ten Sinkhorn iterations, wraps complete encoder layers, and returns to one stream by a project-specific uniform mean readout. These choices adapt the reference mHC design rather than reproduce its complete large-model implementation. Section 5.2 reports mixed, probe-dependent evidence for this component.

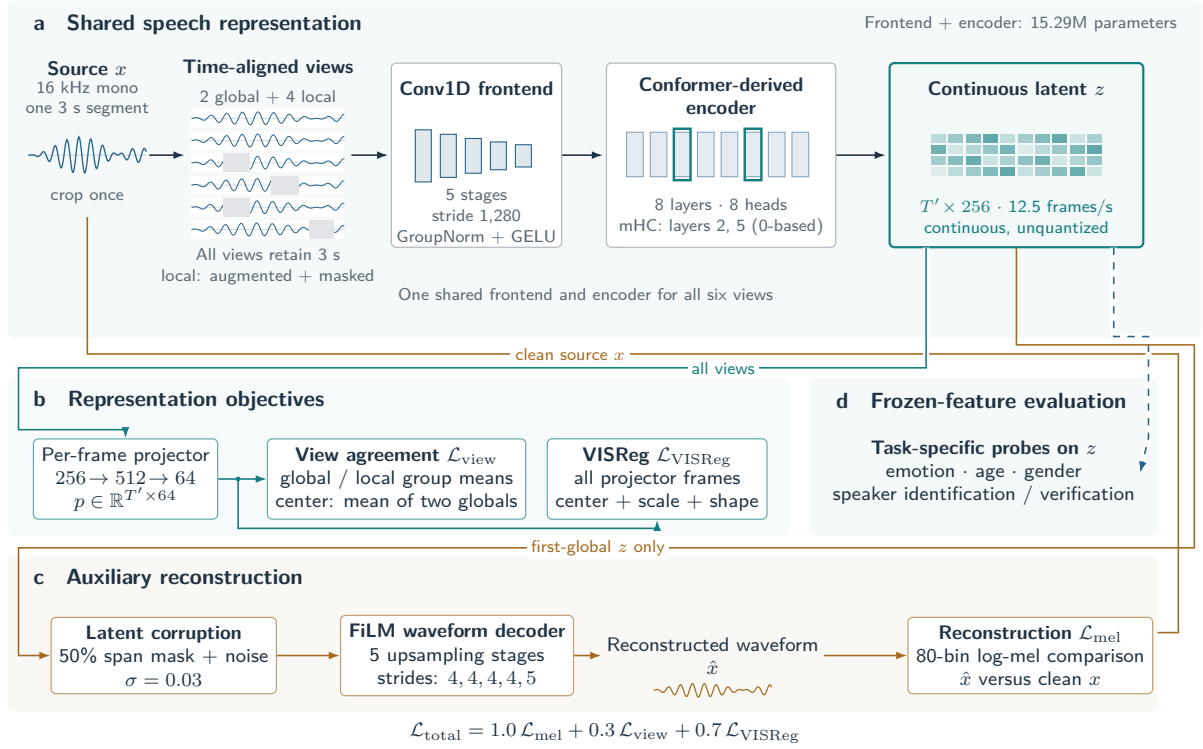


Figure 1: CLAE architecture and objectives. (a) Shared feature extraction produces a continuous 256-dimensional latent at 12.5 Hz; outlined encoder blocks mark the two mHC layers. (b) Representation objectives operate in the 64-dimensional projector space. (c) Auxiliary reconstruction preserves information in the encoder latent. (d) Dashed arrows indicate frozen-feature evaluation. Two global and four full-length local views of the same pre-cropped segment share the frontend, encoder, and projector. Local views use stronger timing-preserving augmentation and frame-aligned masking, not shorter temporal crops. The JEPA-inspired non-contrastive per-frame agreement and supporting VISReg constraint act on projector outputs p ; decoder corruption and waveform reconstruction use the first-global encoder latent z , and frozen-feature probes consume z . Waveforms and feature tiles are schematic.

4 Data and Training

The training manifests combine four Bengali speech sources, with 1,310,076 utterances and close to 2,000 total hours of audio: Common Voice (Ardila et al., 2020), OpenSLR-53 (Kjartansson et al., 2018), RegSpeech12 (Hassan et al., 2025), and Shrutilipi (Bhogle et al., 2023). Table 1 gives the source inventory. The manifests use a 95/5 stratified train/validation split with seed 42; Table 2 gives the resulting counts. No exact audited unique duration is available. Common Voice and OpenSLR-53 both contribute to pretraining, so evaluations on those sources are in-domain frozen-feature evaluations with unassessed train-evaluation overlap. The inventory does not establish demographic or dialect balance, and no deduplication or train-evaluation leakage audit was performed. SUBESCO (Sultana et al., 2021), used for emotion evaluation, is absent from the documented training inventory.

We report the terminal checkpoint after 210,000

Dataset	Utterances
Common Voice Bengali	1,052,178
OpenSLR-53	218,703
regspeech12	21,313
shrutilipi	17,882
Total	1,310,076

Table 1: Bengali speech sources in the training manifests.

training steps, without downstream-metric checkpoint selection. Training uses BF16, AdamW, batch size 42, gradient accumulation over four batches, and gradient clipping at 1.0. The learning-rate schedule was adjusted during training: the base learning rate was reduced to 5×10^{-4} at step 50,000, and later continuations used 10^{-4} with a 300,000-step scheduler horizon. Training ended at step 210,000. Thus, the trajectory includes manual schedule changes rather than a single uninterrupted cosine schedule.

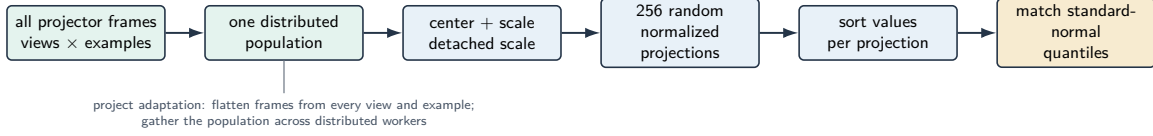


Figure 2: Supporting project-specific VISReg constraint. Projector frames from every view and example are gathered into one distributed population, centered and scaled, projected onto 256 freshly sampled normalized directions, sorted, and matched to standard-normal quantiles.

Split	Utterances
Train	1,244,572
Validation	65,504

Table 2: Stratified 95/5 manifest split (seed 42).

With batch size 42 and four accumulation batches on one worker, the nominal effective batch is 168 source segments. If this batch construction is maintained across the trajectory, 210,000 training steps correspond to 35.28M sampled three-second segments, or 29,400 hours of exposure including repeated sampling. These are exposure estimates, not audited counts of unique audio or successful parameter updates. Each source segment produces six augmented views.

There is no active in-loop validation or validation-based early stopping. The terminal checkpoint is reported without a downstream-metric sweep.

5 Evaluation

We evaluate reconstruction quality and frozen-feature performance on downstream speech tasks. The selected 210k checkpoint is evaluated separately from the matched 25k ablations; their absolute scores should not be interpreted as a checkpoint-matched comparison. Reconstruction metrics are objective proxies. PESQ (Rix et al., 2001), STOI (Taal et al., 2010), ESTOI (Jensen and Taal, 2016), and MR-STFT (Yamamoto et al., 2020) do not substitute for listener judgments.

Appendix A specifies feature extraction, metric definitions, label inventories, and replication constraints. Where available, we compare frozen CLAE features with WavLM (Chen et al., 2022), Whisper (Radford et al., 2023), ECAPA-TDNN (Desplanques et al., 2020), emotion2vec (Ma et al., 2024), Mimi (Défossez et al., 2024), and Higgs Audio V2 tokenizer (Boson AI, 2025). These are

purpose-built systems with different training objectives and specializations; they are reported as task-specific references rather than a homogeneous family of representation models.

5.1 Selected 210k checkpoint

Table 3 reports the main evaluation results. Reconstruction uses a fixed order of 400 three-second Common Voice Bengali validation-manifest clips and FP32 inference. Saved per-item paths identify Common Voice release 26.0 (June 12, 2026); disjointness from pretraining has not been audited. Classification probes keep the frontend and encoder frozen. Emotion, age, and gender use speaker-disjoint folds. Closed-set speaker identification holds out utterances from the same enrolled speakers. Speaker verification uses cosine-scored all-pairs trials, which is a diagnostic rather than a standard enrollment/test benchmark. To keep the reporting consistent with the compact evaluation setting, Table 3 gives the fixed seed-0 point estimate for learned probes; reconstruction and verification have no probe seed.

The table establishes performance only under the stated protocols. Common Voice results are in-domain, and train-evaluation overlap has not been audited. The speaker-identification result is closed-set and does not imply open-set verification performance. The age and gender probes measure label prediction under their speaker-disjoint splits; they do not evaluate fairness or population coverage. The age distribution is highly imbalanced: only two clips have the oldest observed label, and some folds lack rare labels. The 210k age macro-F1 (0.216) is slightly below a training-fold majority-label reference (0.217), so this probe does not establish useful age prediction beyond that reference.

Table 4 places the 210k encoder against frozen external representations. CLAE improves over the architecture-matched random control on these five tasks, but most pretrained baselines have higher

Task	Data and protocol	Metric	Result
Reconstruction	Common Voice, fixed 400 clips	MR-STFT ↓	0.747
		STOI ↑ / ESTOI ↑	0.850 / 0.762
		PESQ-WB ↑	2.146
Emotion (linear)	SUBESCO, 5 speaker-group folds, 7,000 utt.	macro-F1 ↑	0.482
Emotion (temp. attention)	SUBESCO, 1,708 train / 392 held-out utt.	macro-F1 ↑	0.359
Emotion (temp. Transformer)	SUBESCO, 4 speaker-group folds, 7,000 utt.	macro-F1 ↑	0.516
Age	Common Voice, 5 speaker-group folds, 15,475 utt.	balanced acc. / macro-F1 ↑	0.259 / 0.216
Gender	Common Voice, 5 speaker-group folds, 15,690 utt.	balanced acc. / macro-F1 ↑	0.954 / 0.928
Closed-set speaker ID	167 speakers, 666 train / 334 test utt.	accuracy ↑	0.656
Speaker verification	334 speakers, 2,000 utt., all-pairs, mean+std	EER / minDCF ↓	0.276 / 0.988

Table 3: Results for the selected 210k checkpoint. “Temp.” denotes temporal attention or the temporal Transformer, respectively. Probe values are fixed seed-0 point estimates. The tasks use different datasets and protocols and should not be combined into an overall ranking.

Features	Emotion F1 ↑	Age F1 ↑	Gender F1 ↑	Speaker-ID acc. ↑	Verification EER ↓
CLAE, selected 210k	0.482	0.216	0.928	0.656	0.276
Majority label	—	0.217	0.438	—	—
Random CLAE	0.305	0.168	0.742	0.039	0.391
WavLM	0.602	0.249	0.935	0.296	0.339
Whisper-tiny	0.634	0.234	0.944	0.356	0.375
ECAPA	0.519	0.251	0.973	0.952	0.060
emotion2vec	0.629	0.259	0.894	0.326	0.344
Mimi	0.610	0.257	0.957	0.647	0.231
Higgs Audio V2	0.452	0.235	0.928	0.725	0.186

Table 4: Full-data references for the 210k model. The majority-label predictor uses the most frequent training-fold label for each demographic task; dashes mark tasks not evaluated with this reference. Verification uses mean+standard-deviation pooling and the nonstandard all-pairs protocol. Models and tasks have different specializations; the columns should not be averaged into a composite ranking.

emotion and demographic F1, and ECAPA is stronger on speaker identification and verification. These results support task-specific utility rather than broad superiority. Additional full-data probe aggregates and baseline diagnostics appear in Tables 8, 9, and 10. Training-loss diagnostics for the six matched conditions appear in Figure 3.

5.2 Matched 25k ablations

Table 5 compares model variants under a fixed training budget in the earlier matched evaluation. All model conditions stop at 25,000 steps while retaining the base configuration’s 100,000-step learning-rate horizon. Available reconstruction aggregates use a common fixed-order, 400-clip validation-manifest protocol in FP32. Corpus-membership and pretraining-overlap caveats are the same as for the 210k evaluation. The temporal-attention probe uses 1,708 training and 392 held-out SUBESCO utterances. The temporal Transformer uses four speaker-group folds over 7,000 SUBESCO clips; it is a different probe family, not another seed of the attention probe. Each temporal result is a single-seed point estimate.

At this budget, removing reconstruction substantially worsens all four reported reconstruction metrics and both temporal-emotion probe results. This shows the effect of removing the training signal in this setting; it does not establish that z collapsed or that reconstruction is universally necessary. The 25 Hz condition improves STOI, ESTOI, PESQ-WB, and temporal-Transformer F1 relative to the 12.5 Hz base, while doubling the latent frame rate. It is therefore not a matched representation-rate or bitrate comparison.

The mHC result is probe-dependent. Removing mHC slightly lowers MR-STFT and raises temporal-Transformer F1, whereas retaining mHC raises temporal-attention F1. The available evidence does not support a uniform benefit. Full objective and the separately pretrained Mimi reference exhibit a metric-dependent trade-off on common evaluation inputs: the full-objective model is lower on MR-STFT and higher on ESTOI; Mimi is marginally higher on STOI and PESQ-WB. Neither dominates across the reported metrics. The earlier matched evaluation lacks reconstruction-only results and records a failed no-decoder-corruption

Condition	MR-STFT ↓	STOI ↑	ESTOI ↑	PESQ-WB ↑	Attn. F1 ↑	Transf. F1 ↑
Full objective, 25k	0.850	0.828	0.715	1.970	0.341	0.441
Representation-only, 25k	5.652	0.319	-0.001	1.103	0.132	0.197
No mHC, 25k	0.829	0.824	0.721	1.977	0.289	0.449
25 Hz, 25k	0.838	0.857	0.767	2.090	0.318	0.458
No decoder corruption, 25k	—	—	—	—	0.311	0.450
Mimi, 8 quantizers	0.869	0.829	0.707	2.007	n/a	n/a

Table 5: Matched 25k-step ablations and the Mimi common-evaluation reconstruction reference. “Attn.” denotes the temporal-attention probe and “Transf.” the temporal-Transformer probe. Dashes denote the failed reconstruction task in this earlier study. Mimi is separately pretrained, fixes eight quantizers (nominal 1.1 kbps), and is evaluated here only on the common reconstruction inputs.

reconstruction task. Its missing measurement is not replaced by the later full-suite result in Table 8. A separate reduced fixed-subset FP32 comparison includes reconstruction-only (Section 5.3) but uses different sample budgets and must not be merged with these rows.

5.3 Reduced fixed-subset comparison

Table 6 reports a separate reduced study; its scores do not replace or extend the preceding full-data rows. It uses one fixed seed (data, split, and probe seed 0 where applicable) and FP32 CLAE inference with CUDA autotune disabled. Reconstruction uses 100 three-second clips; generic and temporal emotion use the same 1,400 SUBESCO clips, with 20 training epochs for the temporal probes; age and gender each use 2,000 Common Voice clips; speaker identification uses 500 OpenSLR-53 clips; and verification uses 800 OpenSLR-53 clips with 20,000 fixed trials. ASR was deliberately not evaluated. All values are descriptive one-seed outcomes without confidence intervals, error bars, significance tests, or claims of robustness.

CLAE at 210k steps has the lowest STFT loss, while the 25k no-decoder-corruption condition leads PESQ, STOI, age, speaker identification, and both temporal-emotion probes among CLAE conditions. In the direct reduced comparison, reconstruction-only leads generic emotion descriptively (43.89 vs. 41.59 macro-F1), whereas the full-objective model is slightly higher on age (26.11 vs. 25.89) and the two temporal probes (35.21 vs. 34.97 and 42.30 vs. 41.13). These one-seed differences are small and mixed, so they do not demonstrate a uniform benefit of the view-consistency-plus-VISReg branch. The 25 Hz condition leads gender and verification. All CLAE age macro-F1 values in this reduced study are below the majority-label reference (27.42%); this limits any claim of

demographic utility. Representation-only is a negative reconstruction control and also trails the other trained CLAE conditions on every representation task in this reduced protocol.

Whisper-tiny leads generic emotion, Mimi leads age, and ECAPA leads gender, closed-set speaker identification, and verification. WavLM has the strongest configured temporal-Transformer baseline. The eight-quantizer Mimi reconstruction reference obtains STFT 0.8949 on the same 100 clips. Table 11 reports the external-reference scores for this protocol.

6 Limitations

Training did not use validation-based model selection. The main 210k probe results and the matched 25k temporal ablations use one probe seed. We report no paired-bootstrap analysis, confidence intervals, statistical significance, stability, or robustness claims. No human listening study was conducted, so reconstruction evidence is limited to objective proxy metrics. The saved SI-SDR measurements are unexpectedly low (about -23 to -25 dB for reconstruction-capable CLAE conditions in the reduced study, versus $+2.05$ dB for Mimi). Their disagreement with the spectral and intelligibility proxies remains unresolved; we withhold SI-SDR from the comparison tables pending a waveform-alignment and phase-fidelity investigation. The remaining metrics do not establish sample-level waveform fidelity. We did not compare alternative distribution regularizers under a controlled protocol.

The training inventory was not audited for demographic or dialect coverage, deduplication, or leakage. Common Voice and OpenSLR evaluations are in-domain because both sources contribute to pretraining, and their train–evaluation overlap is unknown. Demographic probes predict dataset-

Condition	STFT ↓	PESQ ↑	STOI ↑	Emo. F1 ↑	Age F1 ↑	Gender F1 ↑	Spk-ID acc. ↑	Temp. F1 ↑	Transf. F1 ↑	EER ↓
Full objective, 25k	0.8679	1.983	0.837	41.59	26.11	93.23	67.66	35.21	42.30	27.91
Reconstruction only, 25k	0.8705	2.052	0.842	43.89	25.89	93.32	68.26	34.97	41.13	26.50
Representation only, 25k	4.2770	1.115	0.323	30.21	21.03	90.59	25.75	17.57	19.13	38.92
No mHC, 25k	0.8560	2.013	0.835	41.15	24.69	94.13	69.46	30.29	39.55	27.51
25 Hz, 25k	0.8419	2.139	0.865	43.74	24.77	95.08	70.66	32.41	43.37	23.84
No decoder corruption, 25k	0.7888	2.336	0.882	43.27	27.19	93.24	73.65	36.72	43.56	30.97
CLAE, 210k	0.7745	2.167	0.859	43.21	24.16	94.60	65.87	35.59	42.42	27.11

Table 6: Reduced fixed-subset FP32 CLAE evaluation. Classification scores and EER are percentages. Bold marks the best descriptive CLAE result per column, not a statistically established winner; external and majority-label references are reported separately. ASR was not evaluated.

supplied labels rather than independently verified age or gender. The exact corpus releases and training–evaluation overlap remain insufficiently documented, limiting interpretation beyond these samples. The all-pairs speaker-verification protocol is not a standard enrollment/test benchmark. We draw no conclusion about linguistic retention. The 25 Hz ablation changes frame rate, and mHC effects are mixed. VISReg and mHC are recent preprints used through project-specific adaptations; the reported findings concern the exact local implementation and configuration.

7 Ethical Considerations

Speech can identify speakers and reveal sensitive attributes. The age and gender probes measure predictability of corpus labels; they do not validate those labels as a complete account of identity or justify individual-level decisions. Speaker representations can enable unauthorized identification, surveillance, or impersonation. These research evaluations do not establish suitability for deployment in such settings. Bengali speech and acted-emotion results also should not be generalized to unrepresented dialects, populations, or spontaneous emotion.

The study uses third-party speech corpora. Their availability does not by itself establish consent for demographic inference or speaker identification. The corpus-specific consent terms, release versions, and redistribution permissions require verification before any audio redistribution; this paper makes no claim that those permissions have been independently audited. No new human listening study or participant recruitment was conducted.

Availability. We plan to release the source code, evaluation recipes, and model checkpoints through a public repository. Dataset manifests will not be released. Third-party audio must be obtained from its providers under the applicable terms. The absence

of released manifests will limit exact reconstruction of the sampled subsets; the paper therefore records sample counts, label inventories, and split fingerprints where available.

8 Conclusion

CLAE is a compact 12.5 Hz continuous Bengali speech representation encoder trained with reconstruction as an auxiliary self-supervised information-retention objective, non-contrastive global/local view consistency for nuisance invariance, and frame-level VISReg for anti-collapse and distribution regularization. Its frozen features support emotion and speaker discrimination under the evaluated protocols, while the age probe does not outperform the majority-label reference in macro-F1. The matched 25k study shows a setting-specific benefit from retaining reconstruction, a frame-rate trade-off at 25 Hz, mixed mHC effects, and a metric-dependent comparison with Mimi. The reconstruction-only comparison gives mixed evidence for adding the view-consistency and VIS-Reg branch. CLAE remains continuous and is not bitrate-comparable to quantized codecs. Broader task coverage and repeated-training experiments are needed to assess generality.

References

- Rosana Ardila, Megan Branson, Kelly Davis, Michael Henretty, Michael Kohler, Josh Meyer, Reuben Morais, Lindsay Saunders, Francis M. Tyers, and Gregor Weber. 2020. [Common voice: A massively-multilingual speech corpus](#). In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 4218–4222.
- Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. 2023.

- Self-supervised learning from images with a joint-embedding predictive architecture. *arXiv preprint arXiv:2301.08243*.
- Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. *wav2vec 2.0: A framework for self-supervised learning of speech representations*. In *Advances in Neural Information Processing Systems*, volume 33, pages 12449–12460.
- Randall Balestrieri and Yann LeCun. 2025. *LeJEPa: Provable and scalable self-supervised learning without the heuristics*. *arXiv preprint arXiv:2511.08544*.
- Adrien Bardes, Jean Ponce, and Yann LeCun. 2022. *VICReg: Variance-invariance-covariance regularization for self-supervised learning*. In *International Conference on Learning Representations*.
- Kaushal Santosh Bhogale, Abhigyan Raman, Tahir Javed, Sumanth Doddapaneni, Anoop Kunchukuttan, Pratyush Kumar, and Mitesh M. Khapra. 2023. *Effectiveness of mining audio and text pairs from public data for improving ASR systems for low-resource languages*. In *IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 1–5.
- Boson AI. 2025. Higgs audio v2 tokenizer. Hugging Face model card, <https://huggingface.co/bosonai/higgs-audio-v2-tokenizer>. Accessed: 2026-08-29.
- Edresson Casanova, Daniel Pitol, Wen-Hung Shih, Shreyan Padhy, Sang-gil Kim, Yuchen Zhen, Wuyang Li, Marcos Dourado, Reda Dehak, and Bryan Catanzaro. 2025. *Nanocodec: A low frame rate and lightweight neural audio codec*. In *Interspeech*.
- Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Jian Wu, Michael Zeng, Xiangzhan Zhan, and Furu Wei. 2022. *WavLM: Large-scale self-supervised pre-training for full stack speech processing*. *IEEE Journal of Selected Topics in Signal Processing*, 16(6):1504–1518.
- Alexandre Défossez, Jade Copet, Gabriel Synnaeve, and Yossi Adi. 2023. *High fidelity neural audio compression*. *Transactions on Machine Learning Research*.
- Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. 2024. *Moshi: A speech-text foundation model for real-time dialogue*. *arXiv preprint arXiv:2410.00037*.
- Brecht Desplanques, Jenthe Thienpondt, and Kris Demuynck. 2020. *ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification*. In *Interspeech*, pages 3830–3834.
- Zhengcong Fei, Mingyuan Fan, and Junshi Huang. 2023. *A-JEPA: Joint-embedding predictive architecture can listen*. *arXiv preprint arXiv:2311.15830*.
- Anmol Gulati, James Qin, Chung-Cheng Chiu, Niki Parmar, Yu Zhang, Jiahui Yu, Wei Han, Shibo Wang, Zhengdong Zhang, Yonghui Wu, and Ruoming Pang. 2020. *Conformer: Convolution-augmented transformer for speech recognition*. In *Interspeech*.
- Md. Rezuwan Hassan, Azmol Hossain, Kanij Fatema, Rubayet Sabbir Faruque, Tanmoy Shome, Ruwad Naswan, Trina Chakraborty, Md. Foriduzzaman Zihad, Tawsif Tashwar Dipto, Nazia Tasnim, Nazmuddoha Ansary, Md. Mehedi Hasan Shawon, Ahmed Imtiaz Humayun, Md. Golam Rabiul Alam, Farig Sadique, and Asif Sushmit. 2025. *Regspeech12: A regional corpus of bengali spontaneous speech across dialects*. *arXiv preprint arXiv:2510.24096*.
- Wei-Ning Hsu, Benjamin Bolte, Yao-Hung Hubert Tsai, Kushal Lakhotia, Ruslan Salakhutdinov, and Abdelrahman Mohamed. 2021. *HuBERT: Self-supervised speech representation learning by masked prediction of hidden units*. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 29:3451–3460.
- Jesper Jensen and Cees H. Taal. 2016. *An algorithm for predicting the intelligibility of speech masked by modulated noise maskers*. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 24(11):2009–2022.

- Oddur Kjartansson, Supheakmungkol Sarin, Knot Pipatsrisawat, Martin Jansche, and Linne Ha. 2018. [Crowd-sourced speech corpora for Javanese, Sundanese, Sinhala, Nepali, and Bangladeshi Bengali](#). In *Proceedings of the 6th International Workshop on Spoken Language Technologies for Under-Resourced Languages*, pages 52–55, Gurugram, India.
- Jiaqi Li, Xiaolong Lin, Zhekai Li, Shixi Huang, Yuancheng Wang, Chaoren Wang, Zhenpeng Zhan, and Zhizheng Wu. 2025. [Dualcodec: A low-frame-rate, semantically-enhanced neural audio codec for speech generation](#). In *Inter-speech*.
- Alexander H. Liu, Sang-gil Lee, Chao-Han Huck Yang, Yuan Gong, Yu-Chiang Frank Wang, James R. Glass, Rafael Valle, and Bryan Catanzaro. 2025a. [UniWav: Towards unified pre-training for speech representation and generation](#). In *International Conference on Learning Representations*.
- Chengwei Liu, Haoyin Yan, Shaofei Xue, Xiaotao Liang, Yinghao Liu, Zheng Xue, Gang Song, and Boyang Zhou. 2025b. [UniTok-Audio: A unified audio generation framework via generative modeling on discrete codec tokens](#). *arXiv preprint arXiv:2510.26372*.
- Ziyang Ma, Zhisheng Zheng, Jiaxin Ye, Jinchao Li, Zhifu Gao, Shiliang Zhang, and Xie Chen. 2024. [emotion2vec: Self-supervised pre-training for speech emotion representation](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 15665–15678.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. [Robust speech recognition via large-scale weak supervision](#). In *Proceedings of the 40th International Conference on Machine Learning*, pages 28492–28518.
- Dima Rekesh, Nithin Rao Koluguri, Samuel Kriman, Somshubra Majumdar, Vahid Noroozi, He Huang, Oleksii Hrinchuk, Krishna Puvvada, Ankur Kumar, Jagadeesh Balam, and Boris Ginsburg. 2023. [Fast conformer with linearly scalable attention for efficient speech recognition](#). *arXiv preprint arXiv:2305.05084*.
- Antony W. Rix, John G. Beerends, Michael P. Hollier, and Andries P. Hekstra. 2001. [Perceptual evaluation of speech quality \(PESQ\)—a new method for speech quality assessment of telephone networks and codecs](#). In *IEEE International Conference on Acoustics, Speech, and Signal Processing*.
- Simon Rouard, Manu Orsini, Axel Roebel, Neil Zeghidour, and Alexandre Défossez. 2025. [Continuous audio language models](#). *arXiv preprint arXiv:2509.06926*.
- Sadia Sultana, M. Shahidur Rahman, M. Reza Selim, and M. Zafar Iqbal. 2021. [SUST bangla emotional speech corpus \(SUBESCO\): An audio-only emotional speech corpus for bangla](#). *PLOS ONE*, 16(4):e0250173.
- Cees H. Taal, Richard C. Hendriks, Richard Heusdens, and Jesper Jensen. 2010. [A short-time objective intelligibility measure for time-frequency weighted noisy speech](#). In *IEEE International Conference on Acoustics, Speech, and Signal Processing*.
- Haiyu Wu, Randall Balestrieri, and Morgan Levine. 2026. [VISReg: Variance-invariance-sketching regularization for JEPa training](#). *arXiv preprint arXiv:2606.02572*.
- Zhenda Xie, Yixuan Wei, Huanqi Cao, Chenggang Zhao, Chengqi Deng, Jiashi Li, Damai Dai, Huazuo Gao, Jiang Chang, Kuai Yu, Liang Zhao, Shangyan Zhou, Zhean Xu, Zhengyan Zhang, Wangding Zeng, Shengding Hu, Yuqing Wang, Jingyang Yuan, Lean Wang, and Wenfeng Liang. 2025. [mHC: Manifold-constrained hyperconnections](#). *arXiv preprint arXiv:2512.24880*.
- Ryuichi Yamamoto, Eunwoo Song, and Jae-Min Kim. 2020. [Parallel WaveGAN: A fast waveform generation model based on generative adversarial networks with multi-resolution spectrogram](#). In *IEEE International Conference on Acoustics, Speech, and Signal Processing*.
- Dongchao Yang, Jinchuan Tian, Xu Tan, Rongjie Huang, Songxiang Liu, Haohan Guo, Xuankai Chang, Jiatong Shi, Sheng Zhao, Jiang Bian, Zhou Zhao, Xixin Wu, and Helen M. Meng. 2024. [UniAudio: Towards universal audio generation with large language models](#). In *Proceedings of the 41st International Conference on Machine Learning*.

Shu-wen Yang, Po-Han Chi, Yung-Sung Chuang,
 Cheng-I Jeff Lai, Kushal Lakhotia, Yist Y. Lin,
 Andy T. Liu, Jiatong Shi, Xuankai Chang, Guan-
 Ting Lin, Tzu-Hsien Huang, Wei-Cheng Tseng,
 Ko-tik Lee, Da-Rong Liu, Zili Huang, Shuyan
 Dong, Shang-Wen Li, Shinji Watanabe, Ab-
 delrahman Mohamed, and Hung-yi Lee. 2021.
[SUPERB: Speech processing universal PERfor-
 mance benchmark](#). In *Interspeech*, pages 1194–
 1198.

Neil Zeghidour, Alejandro Luebs, Ahmed Om-
 ran, Jan Skoglund, and Marco Tagliasacchi.
 2021. [SoundStream: An end-to-end neural au-
 dio codec](#). *IEEE/ACM Transactions on Audio,
 Speech, and Language Processing*, 30:495–507.

A Replication Details

CLAE probes use frozen final encoder features z , without layer aggregation or projector features. Audio is mixed to mono and resampled to 16 kHz. CLAE encodes non-overlapping three-second windows, right-pads the final window, and concatenates their frames. Generic utterance probes pool all returned frames, including final-window padding. Reconstruction instead scores the valid waveform prefix when duration is recorded. The two paths therefore have distinct padding conventions.

A.1 Data selection and labels

The demographic loader reads the first sorted local validated.tsv, retains existing audio with nonempty speaker IDs and raw labels, and shuffles eligible rows with the data seed before applying the utterance cap. It neither merges nor corrects labels. Demographic prediction files omit the Common Voice release. Table 7 preserves the raw spelling fourties. Full and reduced age sets contain different classes.

Raw label	Full	Reduced
Age: teens	1,690	197
twenties	12,739	1,674
thirties	871	113
fourties	173	16
fifties	2	0
Gender: female_feminine	3,169	399
male_masculine	12,521	1,601

Table 7: Gold-label counts in the saved out-of-fold predictions. Full/reduced age sets contain 1,580/665 speakers; gender sets contain 1,583/676 speakers.

Generic emotion uses five speaker-group folds on SUBESCO. The full temporal-attention evaluation uses 2,100 clips with 1,708 training and 392 test items; the reduced version uses 1,400 clips with 1,118 training and 282 test items. Both hold out speakers F_09, F_10, M_09, and M_10. The Transformer uses four speaker-group folds. Full speaker ID has 167 enrolled speakers and two held-out utterances per speaker (666 training, 334 test). The reduced set has 500 clips with the same within-speaker holdout rule. These splits separate probe training from testing, not necessarily pretraining from evaluation.

A.2 Probe and metric definitions

Linear probes standardize features using each training fold and fit logistic regression with $C = 1$ and a 3,000-iteration limit; both age and gender use balanced class weights. Speaker ID uses mean pooling; generic emotion, age, and gender use mean+standard-deviation pooling. Reported classification scores are arithmetic means of per-fold scores. Macro-F1 uses labels present in the fold’s gold labels or predictions, as in the evaluator’s default scoring; balanced accuracy averages recall over gold labels present in that fold. Rare-class absence therefore changes the set of scored labels across folds.

The majority-label reference is computed from saved out-of-fold gold labels without feature extraction or model fitting. For each held-out fold, the most frequent label in the other folds is predicted for every test item, and scores are averaged over folds using the same conventions. The majority is twenties for age and male_masculine for gender in every fold. Full/reduced age balanced accuracy is 0.240/0.300 and macro-F1 is 0.216701/0.274245; gender balanced accuracy is 0.500 in both and macro-F1 is 0.437859/0.443730. Recomputing the saved CLAE predictions under this scoring convention reproduces their reported macro-F1.

Reconstruction runs in FP32 and scores each valid source prefix before averaging over clips. MR-STFT uses FFT sizes [256, 512, 1024, 2048], hop length one quarter of each FFT size, full-length Hann windows, centered analysis, and log-magnitude epsilon 10^{-3} . Spectral-convergence, magnitude, and log-magnitude weights are 0.1, 1, and 1, respectively. STOI and ESTOI use pystoi; wide-band PESQ uses pesq at 16 kHz. The selected reconstruction outputs record successful coverage for every requested metric on all 400 or 100 items. Coverage establishes computation, not perceptual validity; the unresolved SI-SDR issue is discussed in the main text.

Verification uses cosine similarity on mean or mean+standard-deviation embeddings; the tables select mean+standard-deviation pooling. Full evaluation scores all unique pairs of 2,000 utterances from 334 speakers. Reduced evaluation has 800 utterances from 134 speakers, with 1,996 target and 18,004 nontarget trials. The capped sampler preferentially retains target pairs before sampling nontarget pairs; it is not uniform all-pairs sub-

sampling. EER averages false-positive and false-negative rates at the nearest ROC crossing. Normalized minDCF uses target prior 0.01 and unit miss/false-alarm costs.

The attention-ASR implementation uses a two-layer, four-head Transformer decoder ($d = 256$, feed-forward dimension 1,024), batch size 16, and an 8,000-update budget. Candidate audio is duration-validated at 15 seconds, encoded in three-second windows, and concatenated without utterance pooling. The full-data development prediction artifact contains 10,000 items. CER and WER are error ratios, so WER may exceed one. These are fixed-budget development-set diagnostics, not a held-out ASR benchmark.

Temporal recipes use a 300-frame cap; the attentive-statistics head has 128 hidden units and batch size 64. The full-suite recipe specifies 40 attention epochs and 30 Transformer epochs; the reduced study report specifies 20 epochs for both. Not all historical optimizer and stopping settings survive in scalar-result metadata, so implementation defaults alone do not establish an exact historical rerun. Full neural-probe aggregates use probe seeds 0/1/2 with fixed data/split seeds 0; main and reduced comparisons use probe seed 0. Deterministic linear-probe repeats have identical scalar scores and are not treated as independent training replications.

External feature extractors remain frozen: WavLM-base-plus and Whisper-tiny use final hidden states; WavLM is final-layer only, not a learned weighted layer aggregation. ECAPA uses its utterance embedding; emotion2vec uses continuous FunASR frame features; Mimi uses its continuous pre-quantization latent; and Higgs Audio V2 decodes its quantizer output to continuous vectors. No codec probe substitutes integer code IDs for continuous features. The separate Mimi reconstruction comparator fixes eight quantizers (nominal 1.1 kbps), rather than reusing the pre-quantization probe feature.

A.3 Model identities and reproducibility limits

External adapters pin the following Hub revisions: WavLM-base-plus, 4c66d4806a428f2e922ccfa1a962776e232d487b; Whisper-tiny, 169d4a4341b33bc18d8881c4b69c2e104e1cc0af; ECAPA, 0f99f2d0ebe89ac095bcc5903c4dd8f72b367286; emotion2vec, 0c9a3152734f9d7a

7a05b4ee6bfb3c109d288664; Mimi, 89091b3e466eb6a9d11e537bf26b144f194978f7; and bosonai/higgs-audio-v2-tokenizer, 403fbacf2f60caaa102f893fdfabb694619b2417.

The selected checkpoint is identified by the run suffix 210k-tail-1r-1e4 and terminal step 210,000. The matched conditions use step-25,000 checkpoints; historical filenames containing 50k are retained run names, not their evaluated training endpoints. Their configuration endpoint is 25,000 with a 100,000-step scheduler horizon. The training configuration records seed 42, AdamW with $\beta = (0.9, 0.999)$, epsilon 10^{-8} , weight decay 10^{-5} , and 5,000 warmup steps. Full-objective loss weights are 1/0.3/0.7 for reconstruction/view/VISReg; reconstruction-only sets the latter two to zero, and representation-only sets the first to zero. No-mHC disables mHC; 25 Hz changes the frontend/decoder stride product to 640; no-decoder-corruption disables its mask and noise.

The available results identify checkpoints by paths and steps rather than immutable weight digests. Reduced-suite staging filenames differ from full-suite run paths, so exact cross-suite weight identity has not been independently verified. The selected run's separate resolved configuration is absent from the local evidence snapshot. Checkpoint-embedded configuration supports the frozen-feature adapters, but the reconstruction and attention-ASR entry points also require a separate configuration file. Hardware history and a complete pinned historical software environment are not available. These limitations prevent a claim of exact end-to-end reproducibility.

For identifying the demographic partitions, full age/gender split fingerprints are, respectively, ef5a12b66a66c4a61a1f9e5a4b907a7e53f3b08bef5e5147ae607f2e2a2def62 and 5f13ef4da550a470843e5124017092104c06c4cef98db732a05b700d927172c7. Reduced age/gender fingerprints are 08d489be6927530a6fc144df52e514daf6a5598ee03c42a4c908bb5fa41d4daa and cc898ed272c53ec29ed85919bbddd4dada217281507534edef4d55a59905b639. Fingerprints identify the saved partitions but cannot reconstruct unreleased manifests.

B Complementary Results

Model condition	MR-STFT ↓	CER ↓	WER ↓	Attn. F1 ↑	Transf. F1 ↑	EER ↓	minDCF ↓
No decoder corruption, 25k	0.775	0.790 ± 0.020	1.106 ± 0.027	0.301 ± 0.011	0.525 ± 0.007	0.319	0.988
Representation only, 25k	5.781	0.899 ± 0.019	1.103 ± 0.042	0.159 ± 0.024	0.227 ± 0.005	0.387	0.999
CLAE, 210k	0.747	0.795 ± 0.010	1.116 ± 0.014	0.341 ± 0.017	0.508 ± 0.008	0.276	0.988

Table 8: Selected full-data reconstruction, ASR, temporal-emotion, and speaker-verification results. For neural probes, $\pm$ denotes sample SD across probe seeds 0, 1, and 2 with fixed data/split seeds 0; it is not a confidence interval or variability across pretrained encoders. EER and minDCF use mean+standard-deviation pooling. These results cover three model conditions; training budgets are indicated in each row.

Model condition	Emotion acc.	Emotion F1	Age bal. acc.	Age F1	Gender bal. acc.	Gender F1	Speaker-ID acc.
No decoder corruption, 25k	0.482	0.478	0.286	0.222	0.947	0.924	0.713
Representation only, 25k	0.360	0.350	0.231	0.153	0.939	0.889	0.162
CLAE, 210k	0.489	0.482	0.259	0.216	0.954	0.928	0.656

Table 9: Deterministic linear-probe results under the same evaluation protocol. Emotion uses SUBESCO; age and gender use Common Voice Bengali with speaker-disjoint splits; speaker identification is closed-set.

Evaluation	Features	Metric 1	Metric 2	Metric 3	Metric 4
Reconstruction	Mimi, 8 quantizers	MR-STFT 0.869	—	—	—
ASR development	WavLM	CER 0.522	WER 0.905	—	—
Temporal emotion	Random CLAE	Attention acc. 0.283	Attention F1 0.229	Transformer acc. 0.298	Transformer F1 0.284
Temporal emotion	WavLM	Attention acc. 0.413	Attention F1 0.372	Transformer acc. 0.612	Transformer F1 0.610

Table 10: Additional reconstruction, development-set ASR, and temporal-emotion baseline results (probe seed 0). Each row is specific to its stated task and metric.

Features	Emotion F1 ↑	Age F1 ↑	Gender F1 ↑	Spk-ID acc. ↑	Temp. F1 ↑	Transf. F1 ↑	EER ↓
Majority label	—	27.42	44.37	—	—	—	—
Random CLAE	28.85	23.41	67.45	7.78	22.67	26.41	38.33
WavLM	53.36	25.57	94.93	34.13	31.53	47.59	34.19
Whisper-tiny	59.79	27.22	97.16	38.32	—	—	37.93
ECAPA	45.32	30.20	99.09	97.01	—	—	6.32
emotion2vec	55.81	25.12	85.12	36.53	—	—	35.42
Mimi	56.62	30.39	95.85	67.66	—	—	22.85
Higgs Audio V2	42.37	23.80	94.47	74.85	—	—	18.34

Table 11: External representations and a demographic majority-label reference on the reduced fixed subsets. Temporal probes were configured only for random CLAE and WavLM. Dashes mean not evaluated, not zero. Model specializations differ, so columns should not be averaged into a composite ranking.

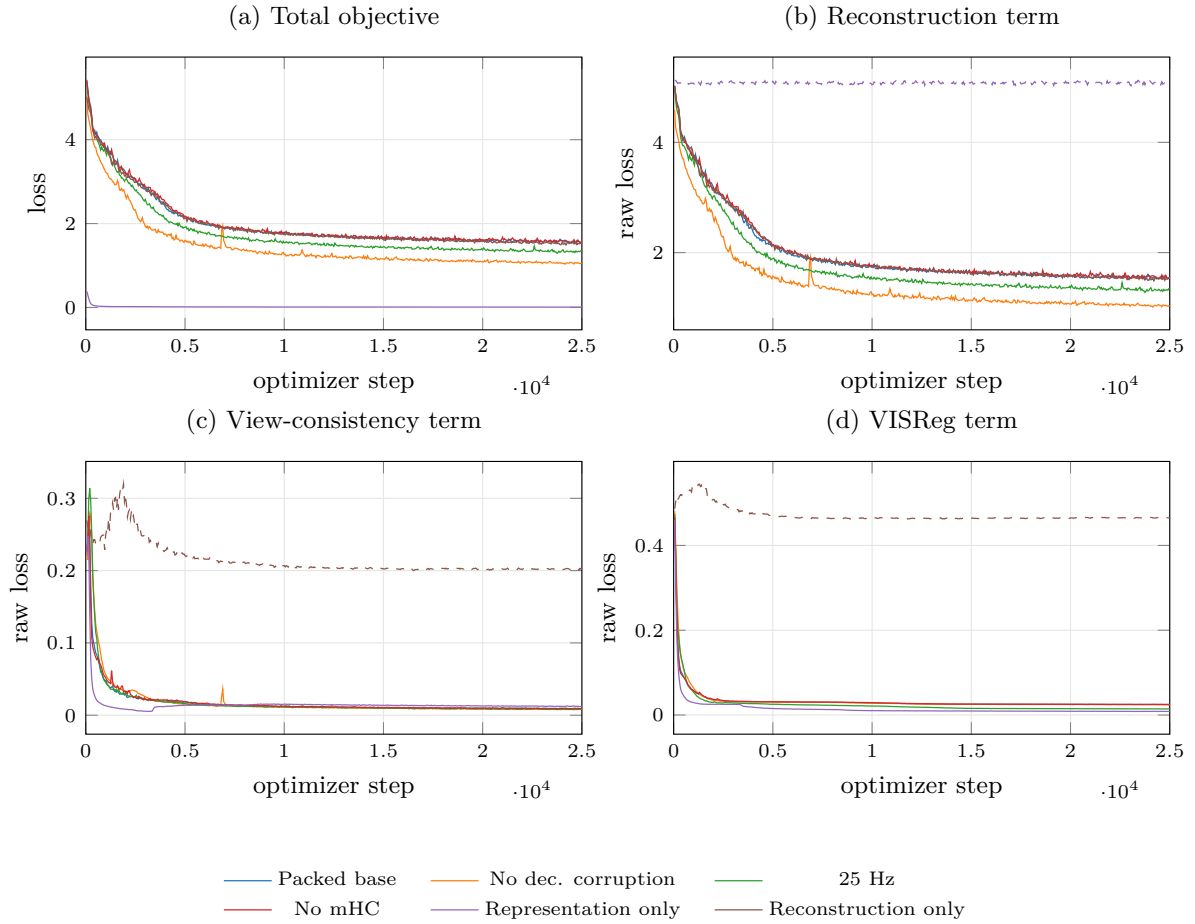


Figure 3: Unsmoothed training loss curves for the six matched 25k conditions. The legend label “Packed base” denotes the full-objective model. Panels (b)–(d) show raw component values, not weighted contributions to the total objective. The dashed reconstruction curve for representation-only and dashed representation diagnostics for reconstruction-only were logged diagnostically but had zero objective weight. These are training diagnostics, not validation curves or evidence of generalization.